

Thought, Action, Observation

DATA 443/643 · Week 14, note 14.2

Week 14

American University

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Ideas from Stanford CS224N 2026 L10 (ReAct vs CoT vs SC). Original slides; that course is not copied.

Lecture plan

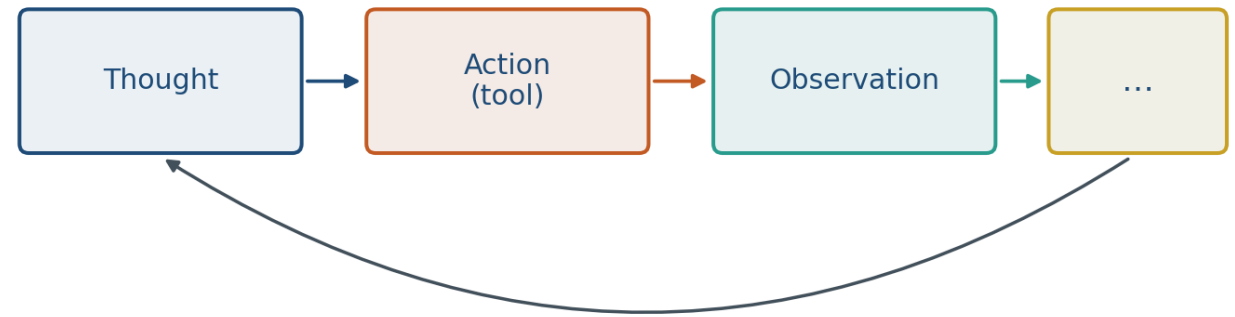
- 1 A tool call is not a thought
- 2 ReAct vs CoT vs self-consistency
- 3 `calc[17*24]` returns 408
- 4 Ignored observations

Key goal: the next thought must use the observation.

Call the world, then continue

- Thought is scratch. Action is structured. Observation comes back.
- CS224N L10: CoT can invent facts. ReAct is supposed to wait for the tool.
- RAG retrieve is one tool. calc is another.

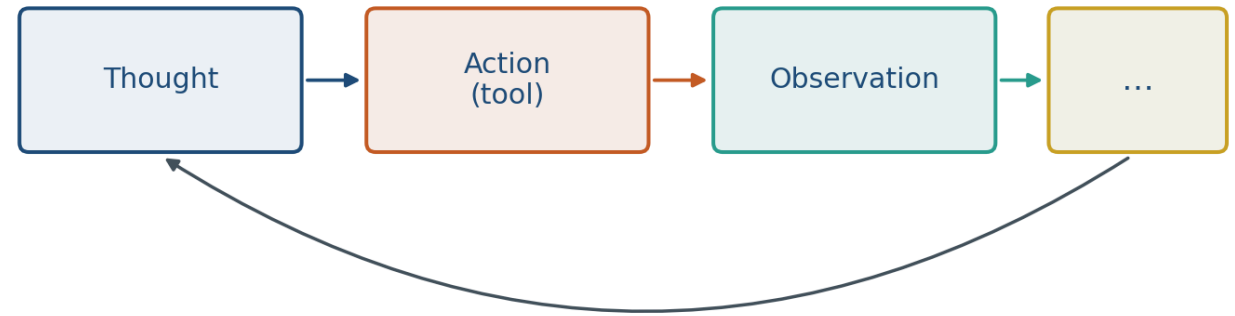
ReAct: think, call a tool, read the result, repeat



Do not multiply in the thought

- Action `calc[17*24]`. Observation 408.
Answer 408.
- If the next thought says 428, the loop is unfaithful.
- Lab 14: `calc[23*60]` → 1380, plus a citation [d1].

ReAct: think, call a tool, read the result, repeat



Three control loops (CS224N L10)

CoT / SC

Text only. SC votes k traces. No observation from the world.

ReAct

Interleave thoughts with tool results. Cap steps. Parse actions strictly.

ToT is still not ReAct

Tree-of-Thoughts

Search among partial thoughts. The scorer can be a checker.

ReAct

The node can be a tool call. Missing a line in the log means you ran CoT.

What to take from this lecture

Five-line log

Thought, Action, Observation,
Thought, Answer.

Dummy calc

Lab 14 is not `python_eval`. Keep it
dummy.

Cap

Looping `calc[1+1]` is a missing step
limit.